THE PERIODIC TABLE 7

The **Periodic Table** is a list of the elements arranged in order of increasing atomic number.

Lavoisier took the first step in classifying elements in 1789. He divided the twenty-five or thirty elements known to him into metals and non-metals. In 1829 Johann Döbereiner noticed that where groups of three similar elements (triads) occurred, the relative atomic mass of the middle element came approximately between those of the other two. However, for many elements the idea did not work. In 1864 John Newlands found that if the elements known to science at the time were arranged in order of increasing relative atomic masses, a pattern emerged. Starting at any element, every eighth element was 'a kind of repetition of the first'. Because of the similarity of this pattern to the musical scale, he called it the **law of octaves**. This 'law' is not very reliable.

1 Explain the difference between *atomic mass* and *atomic number*.

In 1869 Dmitri Mendeléev (or Mendeléeff) made a breakthrough. He arranged the elements in order of increasing relative atomic masses in a similar way to Newlands but, in order to obtain a better repetition, he left gaps for elements which he assumed must exist but had not yet been discovered. He found that many properties then followed a repeating pattern, or **periodic variation**.

Not only did Mendeléev have to leave gaps for undiscovered elements, he also had to correct some relative atomic masses that had been determined erroneously to make the elements fit into the right groups. Mendeléev predicted the existences of 21 missing elements including 'eka-silicon' which is now named germanium. His predicted chemical and physical properties were very close to those actually found when germanium was discovered.

The Periodic Table we now use lists the elements in order of atomic number (not atomic mass). Figure 1 shows an outline of the Periodic Table. There are just over 90 naturally occurring elements.

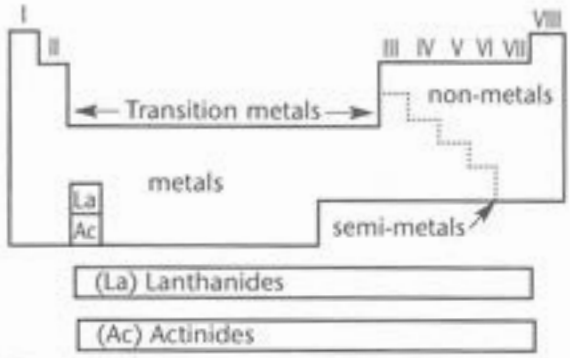


Figure 1

Chemical periodicity refers to the regular occurrence of a set of properties of an element or its compounds in the Periodic Table. Groups are vertical (columns) and are labelled in Roman numerals.

2 Draw up a table showing the Roman numerals from one to eight and their current equivalents.

3 Give two differences between a Periodic Table from Mendeléev's time and a current Periodic Table.

Groups often contain elements with similar properties. **Periods** are horizontal. The Periodic Table can be divided into metals and non-metals. Metallic character is associated with atoms whose nuclei have a low degree of electronic control (i.e. lose electrons in bonding) whereas non-metallic character is associated with atoms whose nuclei have a high degree of control of outer-shell electrons (i.e. gain electrons in bonding).

Some elements exhibit characteristics of both metals and non-metals. These 'semi-metals' or 'metalloids' lie on either side of the dividing line (see the dotted line in Figure 1). Germanium (Ge) and silicon (Si) are examples of these elements.

4 Outline the contributions made by Lavoisier, Döbereiner, Newlands and Mendeléev in establishing the modern Periodic Table.

5 Describe the differences between groups and periods in the Periodic Table.

6 Where in the Periodic Table are the transition metals located?

7 Where in the Periodic Table are the semi-metals located?

8 Outline the properties of semi-metals.

9 Give the names and symbols of two semi-metals.

10 Study a full Periodic Table. What information is given about each element?

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Hydrogen is sometimes placed at the top of Group I metals, but the only thing it has in common with these metals is having one valence electron.

11 What is a *valence electron*?

It takes energy to remove an electron from an atom to form a positive ion. This energy is called **ionisation energy**.

12 The first ionisation of a metal can be represented as $M(s) \rightarrow M^+ + e^-$, where M represents an atom of a metal. Write this representation in words.

13 The second ionisation of a metal can be represented as $M^+ \rightarrow M^{2+} + e^-$. Write this representation in words.

Group I: alkali metals Moving down Group I from lithium (Li) to caesium (Cs), the first ionisation energies decrease (i.e. atoms lose electrons more easily and hence form positive ions more easily) and so reactivity will increase. When Group I metals lose their outer, single valence electron they lose a whole electron shell because the single electron was all that was found in the shell.

Melting points and hardness decrease as you move down the group. The size of the atoms will increase moving down the group because extra electron shells are being added (see table 1).

Element	Atomic radius ($\times 10^{-9}$ m)	Ionic (M^+) radius ($\times 10^{-9}$ m)
Lithium	0.133	0.060
Sodium	0.157	0.095
Potassium	0.203	0.133
Rubidium	0.216	0.148
Caesium	0.235	0.169

Table 1

Alkali metals tend to form ionic compounds with non-metals. Because of their high reactivity, they are not found uncombined in nature. The last element in the group, francium (Fr) is radioactive. Group I metals react with water to form hydrogen and the metal hydroxide as products; e.g. solid sodium (Na) reacts with liquid water (H_2O) to give hydrogen (H_2) gas and a solution of sodium hydroxide (NaOH).

14 Write a balanced formula equation showing the reaction between sodium and water.

15 Give the symbols (showing charge) for the ions formed by the Group 1 metals.

16 Describe what happens to the size of the atomic radius as you move down Group I.

17 Why is the ionic radius for a Group I metal less than its atomic radius?

Group II: alkaline earth metals These elements include beryllium (Be) to radium (Ra) and are all harder and less reactive than the alkali metals. Group II metals exhibit the 2+ oxidation state; i.e. they form ions with a 2+ charge.

18 Give the symbols (showing charge) for the ions formed by the Group II metals.

The Group II metals react with water (but not as strongly as the Group I metals) to form hydrogen and the metal hydroxide; e.g. solid calcium (Ca) reacts with liquid water (H_2O) to give hydrogen (H_2) gas and a solution of calcium hydroxide ($Ca(OH)_2$).

First and second ionisation energies decrease moving down the group from beryllium to radium. As a result, reactivity increases moving down the group. Melting points decrease moving down the group.

19 Give two similarities between Group I and Group II metals.

20 Give two differences between Group I and Group II metals.

Group III Boron (B) is classed as a metalloid, aluminium (Al) is classed as a metal and the other Group III elements are metals. Aluminium does not appear to be reactive because it has a protective oxide coating. Aluminium, gallium (Ga), indium (In) and thallium (Tl) all form 3+ ions.

21 As you move down Group III, what do you expect to happen to the size of the atoms of the individual elements? Explain your answer.

22 Would you expect boron to form ionic or covalent bonds? Explain your answer.

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Group IV This group consists of the elements carbon (C) to lead (Pb). Carbon is a non-metal that forms covalent bonds. Carbon can exist as the allotropes diamond and graphite, which have different properties. Both allotropes contain only the element carbon, but their lattices are arranged differently. In the diamond lattice, all four valence electrons are involved in strong covalent bonds with neighbouring carbon atoms.

Figure 2 shows the layer structure of graphite. Each individual graphite layer is very strongly bonded. Between each layer there are delocalised electrons. Delocalised electrons are not attached to a specific nucleus and they allow graphite to conduct electricity. Graphite is black and much softer than diamond. It can be used as a lubricant as the individual layers slip over each other easily.

Silicon (Si) is a metalloid that forms mainly covalent bonds. Germanium is also a metalloid and forms covalent and ionic (Ge^{4+}) bonds. Tin is a metal and forms ionic (Sn^{2+} and Sn^{4+}) bonds. Lead is also a metal and forms ionic (Pb^{2+} and Pb^{4+}) bonds.

23 As you move down Group IV, what happens to the character of the elements?

24 Describe the structure of graphite.

25 What feature of graphite allows it to conduct electricity?

26 Copy and complete the sentence: *When an element can exist in different forms in the same physical state, each form is called an* _____.

27 Allotropes of the same substance can be very different because of the way their atoms (and electrons) are arranged. Compare the two carbon allotropes, diamond and graphite.

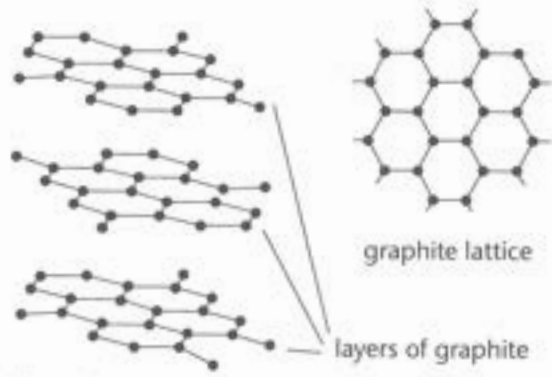


Figure 2

Group V This group has the elements nitrogen (N) to bismuth (Bi). Nitrogen and phosphorus (P), are both definitely non-metals, exhibiting covalent bonding. Nitrogen is a colourless and odourless gas that makes up around four-fifths of the atmosphere. Arsenic (As) and antimony (Sb) are (poisonous) metalloids. Bismuth is a metal. Antimony is a constituent of alloys and semiconductors.

28 Nitrogen exists as diatomic molecules. Explain what is meant by a *diatomic molecule*.

29 Give the formula for a nitrogen molecule.

30 Name the gas that makes up approximately one-fifth of the atmosphere.

31 Explain what is meant by the term *alloy*.

32 Explain what is meant by the term *semiconductor*.

Group VI Oxygen (O) and sulfur (S) are non-metals. The main use of sulfur is making sulfuric acid. Selenium (Se) and tellurium (Te) are metalloids. Selenium is used in the manufacture of solar cells. Polonium (Po), a metal, was the first radioactive element discovered by Marie Curie. A valency of 2 is common in polonium, but others occur.

33 Give the formula for sulfuric acid.

34 Sulfur burns to form the gas sulfur dioxide. Write a balanced formula equation for this reaction.

Group VII The elements are also called halogens ('salt producers'). Their atoms are strong electron acceptors (i.e. strong oxidising agents). Fluorine (F) and chlorine (Cl) are diatomic gases, bromine (Br) is a liquid and iodine (I) is a solid. Their high chemical reactivity makes the halogens corrosive and poisonous. The radioactive element astatine (At) can be made in a nuclear reactor. It resembles iodine but has characteristics that are more metallic. Halogens can form both covalent bonds (with non-metals) and ionic bonds (with metals). Valency is frequently 1 but others can exist.

35 Fluorine reacts with sodium to form sodium fluoride (NaF), a compound used to reduce tooth decay. Write a balanced formula equation for this reaction.

36 Why aren't the halogens found as elements in nature?

37 Seawater contains about 70 parts per million (ppm) by mass of bromine. How much bromine is present in a kilogram of seawater? Show your working.

38 Fluorine is the most reactive halogen. True or false?

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Group VIII This group is made up of the elements helium (He), neon (Ne), argon (Ar), krypton (Kr), xenon (Xe) and radon (Ra). This group is also called the **noble gases**. The atoms of noble gases are stable and other elements tend to react in a way that produces electron configurations (isoelectronic) similar to those of the noble gases. The noble gases are monatomic; i.e. exist as single atoms. The noble gases either do not react (He, Ne, Ar) or react with great difficulty (Kr, Xe, Rn). The compounds formed are unstable. Radon is radioactive.

39 Explain why the noble gases are so unreactive.

40 The noble gases have also been called the *Group 0 elements*. Give a reason for this.

Transition elements These elements are usually quoted as scandium (Sc) to zinc (Zn), yttrium (Y) to cadmium (Cd) and hafnium (Hf) to mercury (Hg). These elements have incomplete inner electron shells and variable valencies. The transition elements have typical metallic properties.

41 Outline three properties of a typical metal.

Many transition elements (and their compounds) act as catalysts; i.e. they speed up reactions without being used up themselves. Transition metal compounds are often coloured. They have strong metallic bonding and are widely used materials because of their mechanical properties.

42 Complete the table with the names and symbols of these transition metals.

Name	Symbol	Name	Symbol	Name	Symbol
Titanium		Iron		Zinc	
	V		Co		Ag
Chromium		Nickel		Gold	
	Mn		Cu		Pt

Lanthanides In the sixth period there is a set of elements from lanthanum (La) to lutetium (Lu). They are also known as the 'rare earth' metals although they are not particularly rare. They are set apart from the main Periodic Table and have very similar chemical and physical properties.

Actinides In the seventh period there is a similar group of elements beginning with actinium (Ac) and ending in lawrencium (Lr). All of the actinide elements are radioactive. Thorium (Th) and uranium (U) occur in nature. The other actinides are manufactured elements (transuranics), produced only as a result of nuclear reactions.

Periodic properties

For the substances listed in the Periodic Table, the properties usually studied include size, melting point, boiling point and first ionisation energy. If these are plotted against atomic number, trends are revealed as you move from left to right across a period.

Size The diameter of atoms increases as you read down a group. This is due to the increasing number of electron shells. As you move across a period, the diameter decreases. This is because there is a greater attraction between the increasing number of protons in the nucleus and the electrons in the same shell.

Structure The changes in melting point that take place across a period are related to structure. If you study a table of melting points, you will see that there are discontinuities (sudden changes); e.g. between carbon and nitrogen and between silicon and phosphorus. This indicates the change in the structure of the elements from giant lattices to discrete molecules.

43 Explain what is meant by a *discrete molecule*.

44 Mark each of the following statements as true or false.

a The lanthanides are very rare elements.

b The actinides are radioactive elements.

c As you move across a period, the size of the atoms increases.

d The lanthanides resemble each other in their properties.

e Elements are grouped on the Periodic Table according to similarities in their properties.

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